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Rethinking the Design Space of Vanilla Pixel Diffusion Transformers

MFlow Learns a Per-Direction Interpolation Between x_0- and Velocity Prediction, Closing the Quality Gap Between Pixel and Latent Diffusion

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Pixel-space diffusion transformers avoid VAE tokenization artifacts but have lagged behind latent diffusion in generation quality, partly because the optimal prediction parameterization—clean image (x_0) versus velocity (u)—depends non-trivially on model capacity and data statistics. MFlow addresses this by introducing per-principal- direction gains that interpolate between x_0- and u-prediction, learned end-to-end with diffusion training. Experiments on ImageNet, UCF-101 video, and large-scale text-to-image generation confirm that MFlow consistently outperforms fixed-parameterization baselines and that larger models systematically prefer velocity prediction. A lightweight ConvRefiner trained post-hoc with an adversarial loss further closes the persistent high-frequency gap.

AT A GLANCE

Problem: The best prediction parameterization for pixel DiTs varies with model capacity and data; the fixed-choice approach leaves quality on the table.

Why it matters: Pixel DiTs promise simpler, higher-fidelity generation than latent diffusion, but their quality gap has blocked adoption.

Method: Per-direction gain $m_i \in [0,1]$ interpolates x_0- and u-prediction; gains are trained jointly with the diffusion model. ConvRefiner handles residual high-frequency deficits post-hoc.

Result: Competitive or improved FID/FVD vs. fixed-parameterization baselines on ImageNet, UCF-101, and text-to-image.

Evidence: Ablations across model scales, architectures, and modalities; systematic m_i analysis validating the capacity- preference relationship.

The Big Picture

Latent diffusion models (LDMs) such as SD3 and FLUX compress images into a low-dimensional latent code before diffusion, enabling fast generation but introducing VAE reconstruction artifacts. Pixel diffusion transformers (pixel DiTs) sidestep this by diffusing directly in pixel space, preserving sharp boundaries and avoiding compression bottlenecks.

The quality gap between pixel DiTs and LDMs has been attributed to less stable training and slower convergence. This paper identifies an underappreciated cause: the prediction parameterization. The choice between predicting the clean image (x_0), the noise (ε), or the velocity ($u = \alpha x_0 - \sigma\varepsilon$) determines the gradient signal across noise levels and model capacity in ways that have not been studied systematically for pixel DiTs.

Why Existing Approaches Fall Short

Fixed x_0-prediction (JiT): JiT showed that x_0-prediction outperforms ε-prediction for pixel DiTs at moderate scale, but this advantage diminishes or reverses at larger model sizes. The fixed choice becomes suboptimal as models scale.

Fixed u-prediction (flow matching): Velocity (flow matching) prediction works well for latent diffusion but underperforms x_0-prediction at smaller pixel DiT scales, because low-capacity models lack the expressiveness to accurately predict the velocity at all noise levels.

AsymFlow: A binary rank-asymmetric assignment of x_0 vs. u prediction to different principal components is a special case of MFlow with step-function gains, and loses the benefit of smooth interpolation.

Core Mechanism

MFlow parameterizes the prediction target for the i-th principal direction of the data as:

$$\hat{y}_i = m_i \cdot u_i + (1 - m_i) \cdot (x_0)_i$$

where the scalar gain $m_i \in [0,1]$ is the sole additional parameter per direction. Setting $m_i = 0$ recovers clean-image prediction; $m_i = 1$ recovers velocity prediction. Gains are initialized from empirical signal-to-noise statistics and updated with gradient descent as part of the standard diffusion training loop.

ConvRefiner is a lightweight CNN trained to predict the residual between a generated pixel DiT image and the real image in the frequency domain. It is trained post-hoc with a spectral adversarial loss while keeping the diffusion backbone frozen, adding only modest compute.

Technical Formulation

Let $x_0 \in \mathbb{R}^d$ be a clean image in pixel space. The flow matching diffusion process defines a noisy image:

$$x_t = \alpha_t x_0 + \sigma_t \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I)$$

The velocity target is $u = \alpha_t x_0 - \sigma_t \varepsilon$. Under MFlow, the model f_θ predicts:

$$\hat{y}_i(x_t, t) = m_i f_\theta^{(u)}(x_t, t)_i + (1 - m_i) f_\theta^{(x_0)}(x_t, t)_i$$

The training loss is the standard flow-matching objective:

$$\mathcal{L}_{\text{MFlow}} = \mathbb{E}_{t, x_0, \varepsilon}\left[\sum_i \left\|\hat{y}_i(x_t, t) - y_i^{\text{target}}\right\|^2\right]$$

where $y_i^{\text{target}} = m_i u_i + (1 - m_i)(x_0)_i$. Gradients flow through both f_θ and $\{m_i\}$.

The ConvRefiner adds:

$$\hat{x}_{\text{refined}} = G_{\text{ref}}(\hat{x}_0) + \hat{x}_0$$

with G_{ref} trained to minimize a spectral GAN loss between $\hat{x}_{\text{refined}}$ and real images.

Learning or Inference Procedure

Training: Standard diffusion training with an extra gradient term for $\{m_i\}$ at each step. The gains are initialised as $m_i = \text{SNR}_i/(1 + \text{SNR}_i)$ where SNR_i is the signal-to-noise ratio along principal component i at the median noise level. ConvRefiner is trained in a second stage with the backbone frozen.

Inference: Identical to standard flow-matching sampling (DDIM, DPM-Solver, or Euler method). The gain m_i does not affect inference speed. ConvRefiner adds one forward pass with the lightweight CNN after the diffusion chain.

What the Guarantee Says

The MFlow loss is a strict generalization of fixed x_0- and u-prediction: setting all $m_i = 0$ recovers the x_0 loss, setting all $m_i = 1$ recovers the velocity loss. Therefore, MFlow can only improve on either fixed choice—its optima weakly dominate—assuming sufficient optimization.

For capacity scaling, the paper proves that for a linear model, the optimal m_i^* satisfies:

$$m_i^* = \frac{\text{Var}[u_i]}{\text{Var}[u_i] + \text{Var}[(x_0)_i]}$$

which is an increasing function of the intrinsic dimensionality of the i-th component—larger models, which can represent more directions, benefit more from velocity prediction.

Experimental Findings

- **ImageNet 256×256:** MFlow reduces FID by 15–20% over fixed x_0- and u-prediction baselines at the same model scale.
- **UCF-101 video:** Consistent FVD improvement across all model sizes; gains are largest for the largest model (+18% over velocity-only baseline).
- **Large-scale text-to-image:** MFlow at 1B parameters matches or exceeds the prior best pixel DiT at 3B parameters.
- **ConvRefiner:** Adds 1.2 ms per image and reduces high-frequency FID deficit by 40% without changing global FID.
- **Capacity scaling:** Empirical m_i values shift from near-zero to near-one as model size increases, confirming the theoretical prediction.

Ablations and Interpretation

Gain initialization: Random initialization of m_i converges to similar final values but requires 2× more training steps; SNR-based initialization is strongly recommended.

Per-direction vs. global gain: A single shared gain m for all directions reduces FID by only 60% of the per-direction improvement, confirming that different principal components benefit differently.

ConvRefiner training signal: Replacing the spectral GAN loss with an ℓ_2 loss in pixel space fails to recover high-frequency detail—the adversarial objective is essential.

Reference

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